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TABLE OF RELATIVE ATOMIC WEIGHTS, 1961

[Reprinted, from the Report of the Commission on Atomic Weights (1961) in the Comptes rendus of the 21st Conference of the International Union of Pure and Applied Chemistry, 1961, with acknowledgement to the International Union of Pure and Applied Chemistry and Butterworths, London.]

BASED ON THE ATOMIC MASS OF $^{12}\text{C} = 12.0$

The values for atomic weights given in the Table apply to elements as they exist in nature, without artificial alteration of their isotopic composition, and, further, to natural mixtures that do not include isotopes of radiogenic origin.

Alphabetical order.

Name	Symbol	Atomic No.	Atomic weight	Name	Symbol	Atomic No.	Atomic weight
Actinium	Ac	89	—	Mercury	Hg	80	200.59
Aluminium	Al	13	26.9815	Molybdenum	Mo	42	95.94
Americium	Am	95	—	Neodymium	Nd	60	144.24
Antimony	Sb	51	121.75	Neon	Ne	10	20.183
Argon	Ar	18	39.948	Neptunium	Np	93	—
Arsenic	As	33	74.9216	Nickel	Ni	28	58.71
Astatine	At	85	—	Niobium	Nb	41	92.906
Barium	Ba	56	137.34	Nitrogen	N	7	14.0067
Berkelium	Bk	97	—	Nobelium	No	102	—
Beryllium	Be	4	9.0122	Osmium	Os	76	190.2
Bismuth	Bi	83	208.980	Oxygen	O	8	15.9994†
Boron	B	5	10.811†	Palladium	Pd	46	106.4
Bromine	Br	35	79.909‡	Phosphorus	P	15	30.9738
Cadmium	Cd	48	112.40	Platinum	Pt	78	195.09
Cæsium	Cs	55	132.905	Plutonium	Pu	94	—
Calcium	Ca	20	40.08	Polonium	Po	84	—
Californium	Cf	98	—	Potassium	K	19	39.102
Carbon	C	6	12.01115†	Praseodymium	Pr	59	140.907
Cerium	Ce	58	140.12	Promethium	Pm	61	—
Chlorine	Cl	17	35.453‡	Protactinium	Pa	91	—
Chromium	Cr	24	51.996‡	Radium	Ra	88	—
Cobalt	Co	27	58.9332	Radon	Rn	86	—
Copper	Cu	29	63.54	Rhenium	Re	75	186.2
Curium	Cm	96	—	Rhodium	Rh	45	102.905
Dysprosium	Dy	66	162.50	Rubidium	Rb	37	85.47
Einsteinium	Es	99	—	Ruthenium	Ru	44	101.07
Erbium	Er	68	167.26	Samarium	Sm	62	150.35
Europium	Eu	63	151.96	Scandium	Sc	21	44.956
Fermium	Fm	100	—	Selenium	Se	34	78.96
Fluorine	F	9	18.9984	Silicon	Si	14	28.086†
Francium	Fr	87	—	Silver	Ag	47	107.870‡
Gadolinium	Gd	64	157.25	Sodium	Na	11	22.9897
Gallium	Ga	31	69.72	Strontium	Sr	38	87.62
Germanium	Ge	32	72.59	Sulphur	S	16	32.064†
Gold	Au	79	196.967	Tantalum	Ta	73	180.948
Hafnium	Hf	72	178.49	Technetium	Tc	43	—
Helium	He	2	4.0026	Tellurium	Te	52	127.60
Holmium	Ho	67	164.930	Terbium	Tb	65	158.924
Hydrogen	H	1	1.00797†	Thallium	Tl	81	204.37
Indium	In	49	114.82	Thorium	Th	90	232.038
Iodine	I	53	126.9044	Thulium	Tm	69	168.934
Iridium	Ir	77	192.2	Tin	Sn	50	118.69
Iron	Fe	26	55.847‡	Titanium	Ti	22	47.90
Krypton	Kr	36	83.80	Tungsten	W	74	183.85
Lanthanum	La	57	138.91	Uranium	U	92	238.03
Lead	Pb	82	207.19	Vanadium	V	23	50.942
Lithium	Li	3	6.939	Xenon	Xe	54	131.30
Lutetium	Lu	71	174.97	Ytterbium	Yb	70	173.04
Magnesium	Mg	12	24.312	Yttrium	Y	39	88.905
Manganese	Mn	25	54.9380	Zinc	Zn	30	65.37
Mendelevium	Md	101	—	Zirconium	Zr	40	91.22

* See Whiffen, *Proc. Chem. Soc.*, 1960, 97.

† Atomic weights so designated are known to be variable because of natural variations in isotopic composition. The observed ranges are:

Hydrogen	± 0.00001	Carbon	± 0.00005	Silicon	± 0.001
Boron	± 0.003	Oxygen	± 0.0001	Sulphur	± 0.003

‡ Atomic weights so designated are believed to have the following experimental uncertainties:

Chlorine	± 0.001	Iron	± 0.003	Silver	± 0.003
Chromium	± 0.001	Bromine	± 0.002		

Reprints of these Tables, on white card, may be obtained from the General Secretary, the Chemical Society, price 2s. each (post free).

TABLE OF RELATIVE ATOMIC WEIGHTS, 1961

BASED ON THE ATOMIC MASS OF $^{12}\text{C} = 12$.

The values for atomic weights given in the Table apply to elements as they exist in nature, without artificial alteration of their isotopic composition, and, further, to natural mixtures that do not include isotopes of radiogenic origin.

Order of atomic number.							
Atomic No.	Name	Symbol	Atomic weight	Atomic No.	Name	Symbol	Atomic weight
1	Hydrogen	H	1.00797 †	52	Tellurium	Te	127.60
2	Helium	He	4.0026	53	Iodine	I	126.9044
3	Lithium	Li	6.939	54	Xenon	Xe	131.30
4	Beryllium	Be	9.0122	55	Cesium	Cs	132.905
5	Boron	B	10.811 †	56	Barium	Ba	137.34
6	Carbon	C	12.01115 †	57	Lanthanum	La	138.91
7	Nitrogen	N	14.0067	58	Cerium	Ce	140.12
8	Oxygen	O	15.9994 †	59	Praseodymium	Pr	140.907
9	Fluorine	F	18.9984	60	Neodymium	Nd	144.24
10	Neon	Ne	20.183	61	Promethium	Pm	—
11	Sodium	Na	22.9898	62	Samarium	Sm	150.35
12	Magnesium	Mg	24.312	63	Europium	Eu	151.96
13	Aluminium	Al	26.9815	64	Gadolinium	Gd	157.25
14	Silicon	Si	28.086 †	65	Terbium	Tb	158.924
15	Phosphorus	P	30.9738	66	Dysprosium	Dy	162.50
16	Sulphur	S	32.064 †	67	Holmium	Ho	164.930
17	Chlorine	Cl	35.453 ‡	68	Erbium	Er	167.26
18	Argon	Ar	39.948	69	Thulium	Tm	168.934
19	Potassium	K	39.102	70	Ytterbium	Yb	173.04
20	Calcium	Ca	40.08	71	Lutetium	Lu	174.97
21	Scandium	Sc	44.956	72	Hafnium	Hf	178.49
22	Titanium	Ti	47.90	73	Tantalum	Ta	180.948
23	Vanadium	V	50.942	74	Tungsten	W	183.85
24	Chromium	Cr	51.996 ‡	75	Rhenium	Re	186.2
25	Manganese	Mn	54.9380	76	Osmium	Os	190.2
26	Iron	Fe	55.847 ‡	77	Iridium	Ir	192.2
27	Cobalt	Co	58.9332	78	Platinum	Pt	195.09
28	Nickel	Ni	58.71	79	Gold	Au	196.967
29	Copper	Cu	63.54	80	Mercury	Hg	200.59
30	Zinc	Zn	65.37	81	Thallium	Tl	204.37
31	Gallium	Ga	69.72	82	Lead	Pb	207.19
32	Germanium	Ge	72.59	83	Bismuth	Bi	208.980
33	Arsenic	As	74.9216	84	Polonium	Po	—
34	Selenium	Se	78.96	85	Astatine	At	—
35	Bromine	Br	79.909 ‡	86	Radon	Rn	—
36	Krypton	Kr	83.80	87	Francium	Fr	—
37	Rubidium	Rb	85.47	88	Radium	Ra	—
38	Strontium	Sr	87.62	89	Actinium	Ac	—
39	Yttrium	Y	88.905	90	Thorium	Th	232.038
40	Zirconium	Zr	91.22	91	Protactinium	Pa	—
41	Niobium	Nb	92.906	92	Uranium	U	238.03
42	Molybdenum	Mo	95.94	93	Neptunium	Np	—
43	Technetium	Tc	—	94	Plutonium	Pu	—
44	Ruthenium	Ru	101.07	95	Americium	Am	—
45	Rhodium	Rh	102.905	96	Curium	Cm	—
46	Palladium	Pd	106.4	97	Berkelium	Bk	—
47	Silver	Ag	107.870 ‡	98	Californium	Cf	—
48	Cadmium	Cd	112.40	99	Einsteinium	Es	—
49	Indium	In	114.82	100	Fermium	Fm	—
50	Tin	Sn	118.69	101	Mendelevium	Md	—
51	Antimony	Sb	121.75	102	Nobelium	No	—

† Atomic weights so designated are known to be variable because of natural variations in isotopic composition. The observed ranges are:

Hydrogen	± 0.00001	Carbon	± 0.00005	Silicon	± 0.001
Boron	± 0.003	Oxygen	± 0.0001	Sulphur	± 0.003

‡ Atomic weights so designated are believed to have the following experimental uncertainties:

Chlorine	± 0.001	Iron	± 0.003	Silver	± 0.003
Chromium	± 0.001	Bromine	± 0.002		

EDITORIAL REPORT ON NOMENCLATURE, 1962.

1. NOMENCLATURE OF CARBOHYDRATES

A revision and slight extension of the U.S.-British Rules of Carbohydrate Nomenclature, 1952 (given in Chapter 5 of "Handbook for Chemical Society Authors, 1961") has been agreed between the appropriate U.S. and British Committees, and accepted by the Chemical Society and the American Chemical Society for use in their respective periodicals. The new rules will form the basis of the chapter on carbohydrate nomenclature in any future revision of the "Handbook"; most of the revisions are verbal, to remove ambiguities, and have no effect on the resulting nomenclature. The following major changes and additions should, however, be noted.

Rule 7 [This rule, describing the naming of ketoses by the use of the termination "ulose," has been extended by the following additions.] A sugar which is both an aldose and a ketose will be named by replacing the final "e" in the name of the (formally) parent aldose by the suffix "-ulose," preceded by the appropriate position numeral, and then adding the appropriate configurational prefixes.

Ketoses having two ketonic carbonyl groups will be named by means of the suffix "-diulose"; before this will be placed a prefix denoting the number of carbon atoms in the chain, which, in turn, will be preceded by the numerals designating the positions of the carbonyl groups and the prefix denoting the configuration of the group of asymmetric centres present (see Rule 6).

Examples:

(a) Permissible trivial names established by usage:

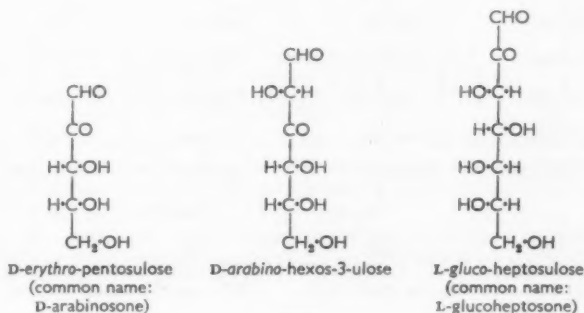
D-fructose for D-arabino-hexulose, D-psicose for D-ribo-hexulose, D-sorbose for D-xylo-hexulose, D-tagatose for D-lyxo-hexulose, and the corresponding names for the L-forms; sedoheptulose for D-altro-heptulose.

(b) Names that cannot be used as definite names, because they have incorrect structural implications:

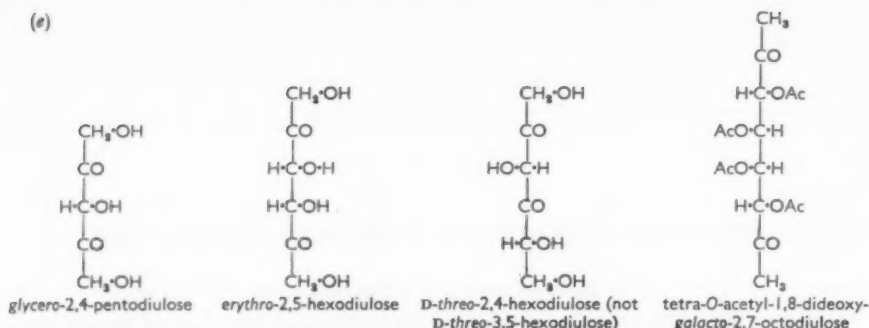
glycerulose (for 1,3-dihydroxy-2-propanone), D-erythrulose or D-threulose (for D-glycero-tetrolulose), D-arabinulose or D-ribulose (for D-erythro-pentulose), D-lyxulose or D-xylulose (for D-threo-pentulose), and the corresponding names for the L-forms.

(c) [This section lists the previously published examples.]

(d)



(e)



Rule 17 [The prefixes *aldehydo* and *keto* are now to be italicized.]

Rule 20 [The nomenclature of glycosides has been restricted by redefinition to:] a glycoside is a mixed acetal resulting from the exchange of an *alkyl* or *aryl* radical for the hydrogen atom of the hemiacetal hydroxyl group of a cyclic form of an aldose or ketose.

Rule 22 [This rule has been re-drafted and is therefore now given in full.] A monosaccharide containing more than four configurational asymmetric carbon atoms will be named by adding two or more prefixes, indicating the configurations of those asymmetric carbon atoms, to a root indicating the number of carbon atoms in the chain and ending with the suffix "-ose" for aldoses and "-ulose" for ketoses.

The configurational prefixes employed are given in Rule 6. For the aldoses and 2-ketoses, the sequence of asymmetric carbon atoms will be divided into units, commencing, with a unit of four asymmetric carbon atoms, at the asymmetric carbon atom next to the functional group (see below). The order of citation of these prefixes will commence at the end farthest from carbon atom number one, and proceed along the carbon chain to the asymmetric carbon atom nearest to carbon atom number one. In designating the anomeric ring forms of these aldoses and 2-ketoses, the anomeric prefix here (see Rule 15) will immediately precede the configurational prefix for that group of asymmetric carbon atoms, next to the functional group, which has one or more atoms involved in the ring.

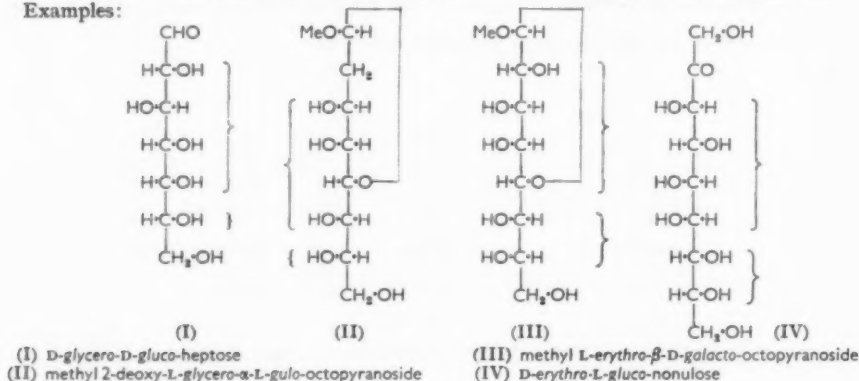
Number of
asymmetric
carbons in
the sequence

Prefixes to be used

	Order of selection
5	one 4-carbon + one 1-carbon
6	one 4-carbon + one 2-carbon
7	one 4-carbon + one 3-carbon
8	two 4-carbon
9	two 4-carbon + one 1-carbon
10	two 4-carbon + one 2-carbon
11, etc.	two 4-carbon + one 3-carbon, etc.

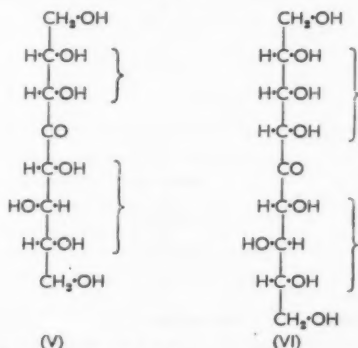
	Order of citation
one 1-carbon + one 4-carbon	
one 2-carbon + one 4-carbon	
one 3-carbon + one 4-carbon	
two 4-carbon	
one 1-carbon + two 4-carbon	
one 2-carbon + two 4-carbon	
one 3-carbon + two 4-carbon, etc.	

Examples:



For other ketoses, the sequence of asymmetric carbon atoms will be similarly divided into units that commence with the group of three (or, if present, four) highest-numbered asymmetric carbon atoms that are next to the functional group.

Examples:

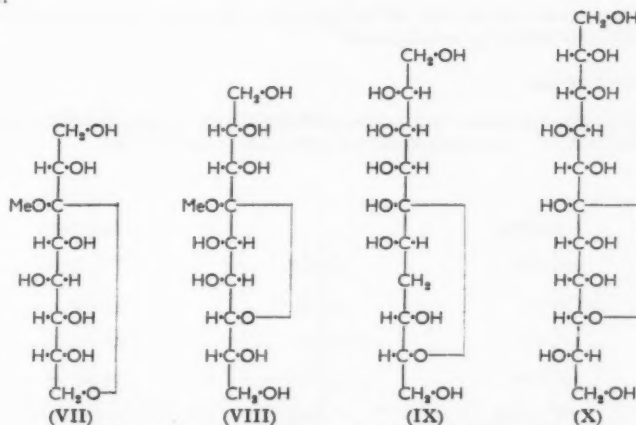


(V) D-xylo-D-erythro-4-octulose

(VI) D-xylo-D-ribo-5-nonulose or L-ribo-L-xylo-5-nonulose

For oxygen-ring forms of ketoses other than 2-ketoses, the ending (furanose, furanoside, etc.) will, when necessary, be immediately preceded by a pair of numerals identifying the two carbon atoms to which the oxygen ring is attached, the potential ketone group being cited first. When the potential ketone group of an oxygen-ring form is in the middle of the chain, the anomeric carbon atom has, as usual, the top vertical orientation, and numbering proceeds downward and toward the ring-forming hydroxyl group.

Examples:



(VII) methyl β-D-glucio-D-glycero-3-octuloseptanoside

(VIII) methyl β-D-manno-D-erythro-4-nonulo-4,7-furanoside

(IX) 7-deoxy-β-D-arabino-L-ribo-5-deculo-5,9-pyranose

(X) α-L-talo-D-gulo-6-undeculo-6,9-furanose

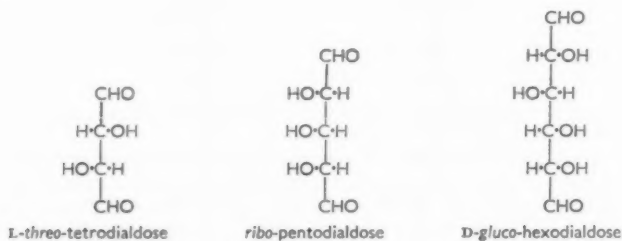
Comment. If the carbonyl group is, with reference to the asymmetric carbon atoms, in the middle of the chain, the sugars resemble the alditols (Rule 23) and dialdoses (Rule 24), in that some are *meso*-forms, for which the names are written without D or L, and some have two equally correct names [see example (VI)].

Rule 23 [Since the list of permissible names of alditols, established by usage, has now been omitted, the names of these compounds should be properly derived; also arabinitol should not be abbreviated to arabitol.]

Rule 24 [This rule, previously under consideration, has now been published.] Dialdehydes formed from aldoses having a terminal CH_2OH group by oxidation of this group only to CHO may be named by attaching the suffix "dialdose" to a root indicating the number of carbon atoms in the chain and then adding prefixes indicating the configurations of the asymmetric carbon atoms.

Examples:

(a)



(b) Names requiring D or L:

threo-tetrodialdose, *arabino*-pentodialdose, *gluco*-hexodialdose, *ido*-hexodialdose, *manno*-hexodialdose, *talo*-hexodialdose.

(c) Names of *meso*-forms are used without D or L. The prefix *meso*- may be inserted for clarity:

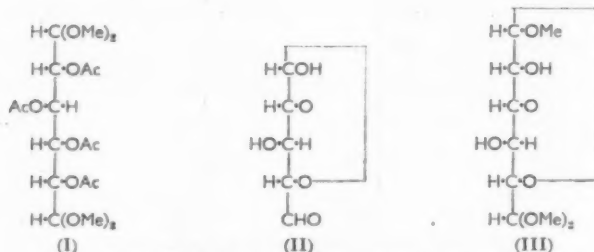
erythro-tetrodialdose, *ribo*-pentodialdose, *xylo*-pentodialdose, *allo*-hexodialdose, *meso-galacto*-hexodialdose.

The names of *meso*-forms can be used advantageously with D or L in naming derivatives which have become optically active by substitution.

(d) Equivalent names:

D-*arabino*-pentodialdose and D-*lyxo*-pentodialdose; D-*gluco*-hexodialdose and L-*gulo*-hexodialdose; D-*talo*-hexodialdose and D-*altro*-hexodialdose.

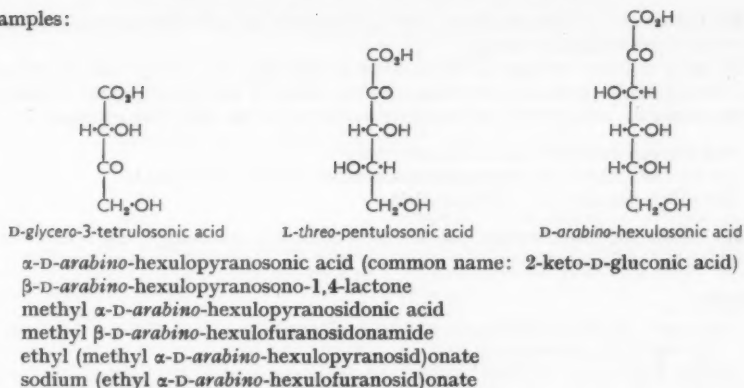
(e)



(I) tetra-O-acetyl-D-gluco-hexodialdose bis(dimethyl acetal)
 (II) α -D-xylo-pentodialdo-1,4-furanose
 (III) methyl α -D-gulo-hexodialdo-1,5-pyranoside 6-(dimethyl acetal)

New Rule 26 [This is a completely new rule.] When a secondary alcoholic hydroxyl group of an aldonic acid is oxidized to a carbonyl group, the compound will be named by means of the suffix "ulosonic acid" attached to a root designating the number of carbon atoms in the chain, which is in turn preceded by the appropriate numeral indicating the position of the ketonic function when other than 2 (see Rule 7). The configuration of the acid will be designated as given in Rule 6.

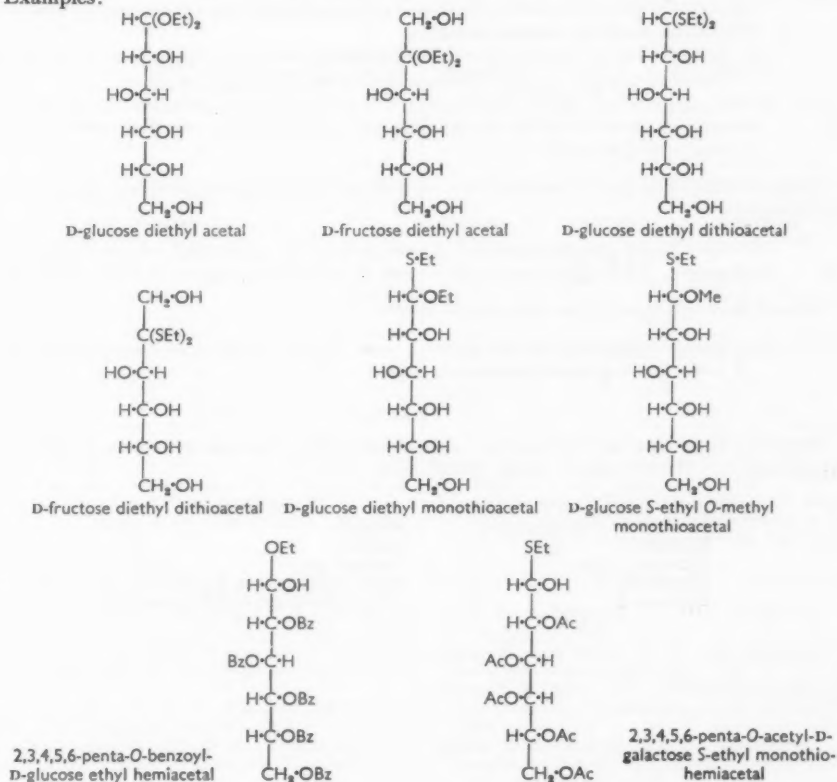
Examples:



[Old Rules 26—30 have been renumbered 27—31.]

Rule 32 [This rule, originally under discussion, is now promulgated.] In naming acetals formed between alcohols (or thiols) and the carbonyl carbon atom of the acyclic forms of aldoses or ketoses, the name of the sugar will be the first word, "acetal" ("dithioacetal") the last word, and the name of the appropriate radical (or radicals) will be the middle word (or words). Derivatives containing the hemiacetal function are named in an analogous manner.

Examples:



Rule 33 [The name "sedoheptulosan" (for 2,7-anhydro- β -D-*allo*-heptulopyranose) is added as a trivial name established by usage.]

Rule 34 [As a result of revision of the wording of this Rule (to accord with the principle of Rule 15), the anomeric prefix now accompanies the name of the sugar residue instead of the appropriate numeral; anhydrides are therefore named as in the following examples.]

di- β -D-fructofuranose 1,2':2,3'-dianhydride
 β -D-fructofuranose β -D-*threo*-pentulofuranose 1,2':2,1'-dianhydride
 di-D-ribofuranose 1,5':1',5-dianhydride

Rule 35. Reducing disaccharides may also now be named according to the system described for tri- and higher oligo-saccharides.

Examples:

α -lactose: 4-O- β -D-galactopyranosyl- α -D-glucopyranose or O- β -D-galactopyranosyl-(1 $\rightarrow$ 4)- α -D-glucopyranose
 methyl α -lactoside: methyl 4-O- β -D-galactopyranosyl- α -D-glucopyranoside or methyl O- β -D-galactopyranosyl-(1 $\rightarrow$ 4)- α -D-glucopyranoside
 2-amino-2-deoxy-4-O-(β -D-galactopyranosyl)- α -D-glucopyranose or O- β -D-galactopyranosyl-(1 $\rightarrow$ 4)-2-amino-2-deoxy- α -D-glucopyranose
 3-O-(4-O-methyl- α -D-galactopyranosyluronic acid)- α -D-glucopyranose or O-(4-O-methyl- α -D-galactopyranosyluronic acid)-(1 $\rightarrow$ 3)- α -D-glucopyranose
 3-O-(2-amino-2-deoxy- β -D-glucopyranosyl)- α -D-glucopyranuronic acid or O-(2-amino-2-deoxy- β -D-glucopyranosyl)-(1 $\rightarrow$ 3)- α -D-glucopyranuronic acid
 6-O-(methyl tri-O-acetyl- β -D-mannopyranosyluronate)-tri-O-acetyl- α -D-altropyranosyl bromide or O-(methyl tri-O-acetyl- β -D-mannopyranosyluronate)-(1 $\rightarrow$ 6)tri-O-acetyl- α -D-altropyranosyl bromide
 4-O-(methyl 4-O-methyl- α -D-glucopyranosyluronate)- α -D-xylopyranose or O-(methyl 4-O-methyl- α -D-glucopyranosyluronate)-(1 $\rightarrow$ 4)- α -D-xylopyranose
 phenyl 2-amino-2-deoxy-3-O-(ethyl β -D-glucopyranosyluronate)- β -D-galactopyranoside or phenyl O-(ethyl β -D-glucopyranosyluronate)-(1 $\rightarrow$ 3)-2-amino-2-deoxy- β -D-galactopyranoside

[Slight modification to the written form of non-reducing oligosaccharides is illustrated by the following.]

raffinose: O- α -D-galactopyranosyl-(1 $\rightarrow$ 6)- α -D-glucopyranosyl β -D-fructofuranoside
 gentianose: O- β -D-glucopyranosyl-(1 $\rightarrow$ 6)- α -D-glucopyranosyl β -D-fructofuranoside

[Whilst the naming of polyuronic acids is shown by]

O-(α -D-galactopyranosyluronic acid)-(1 $\rightarrow$ 4)-O-(α -D-galactopyranosyluronic acid)-(1 $\rightarrow$ 4)- α -D-galactopyranuronic acid

2. VITAMINS

Errors of transcription occur in the table of fat-soluble vitamins given on p. 200 of the "Handbook". The vitamin A group should read:

Customary designation hitherto	Name adopted
Vitamin A ₁ or axerophthol	Retinol
Retinene	Retinal
Vitamin A acid	Retinoic acid
Vitamin A ₂	3-Dehydroretinol or dehydroretinol
Retinene 2	3-Dehydroretinal or dehydroretinal

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